

Causal calibration in xAI

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Abstract

The insertion/deletion tests of Petsiuk et al. (2018) are the most popular evaluation method for saliency maps: pixels are ranked by attributed importance and progressively removed or revealed while the model’s confidence is tracked. We study an apparent issue in this approach: the perturbed images that the test feeds to the model are heavily out of distribution (OOD), so the confidence that the final curve is built from is itself untrustworthy, biasing the area under curve (AUC) score and producing noisy, non-monotone curves. Following the ReCalX framework (Decker et al., 2025), we recalibrate each model with *perturbation-aware temperature scaling*: one temperature per perturbation bin, fitted by negative log likelihood minimisation, applied without changing the model’s predictions. We further propose *significance bucketing*, a grouping approach that removes equal count groups of pixels instead of a fixed top- $n\%$. Across four models (ResNet50/101, ViT-B/16, ViT-B/32) and six attribution methods, calibration substantially reduces miscalibration. On ResNet50, it drops average ECE ($0.28 \rightarrow 0.16$, significant for five of six methods) and TACE ($0.35 \rightarrow 0.20$, significant for all six methods). Monotonicity improvement is method-dependent: significant for LIME and Feature Ablation, absent or reversed for strong gradient methods. Bucketing is significantly more monotone than fixed-step removal across both raw and calibrated settings, improving the baseline Spearman r from -0.754 to -0.811 and the monotonicity ratio from 0.555 to 0.632 (Wilcoxon $p < 0.01$). Together the two upgrades make insertion/deletion a more trustworthy XAI metric.

1. Introduction

Attribution methods explain an image classifier by assigning each input pixel a score for how much it supports a target class. To decide whether such an explanation is accurate, the common approach is to rely on the causal insertion/deletion test (Petsiuk et al., 2018): rank pixels by importance, then delete (or insert) them in order and track the predicted probability. A good map makes confidence collapse quickly under deletion (yielding a low AUC).

However, masking pixels creates out-of-distribution (OOD) data points, causing the classifier to become severely miscalibrated. Consequently, the confidence curve underlying the AUC becomes non-monotone and untrustworthy.

Research questions. (RQ1) Can post-hoc recalibration mitigate the OOD reliability gap and reduce miscalibration during causal tests what will result in more monotone curves? **(RQ2)** Does removing pixels in equal-count significance buckets provide a more monotone evaluation than the standard fixed-step top- $n\%$ removal?

Contributions. (C1) We show that perturbation-binned temperature scaling (ReCalX) significantly reduces miscalibration across 5 of 6 attribution methods on ResNet50 (ECE $0.28 \rightarrow 0.16$, TACE $0.35 \rightarrow 0.20$; Wilcoxon $p < 0.001$), confirming that the OOD gap is a calibration issue fixable post-hoc. **(C2)** We show that bucket deletion yields significantly more monotone curves than fixed-step 1% removal across both Spearman r (from -0.754 to -0.811 , $p < 0.01$) and monotonicity ratio (from 0.555 to 0.632 , $p < 0.01$). **(C3)** We identify that calibration’s effect on monotonicity is highly method-dependent: it provides significant improvements for LIME and Feature Ablation, but the effect is absent or reversed for strong gradient-based methods.

2. Methodology

2.1 Attribution methods: inputs, outputs, meaning

An attribution method takes a trained model f , an input image $x \in \mathbb{R}^{3 \times 224 \times 224}$ and a target class c , and returns a 2D importance map $S \in \mathbb{R}^{224 \times 224}$ where S_{ij} scores how much pixel (i, j) supports $f_c(x)$. We evaluate six methods spanning two families. *Gradient/back-propagation* methods: Saliency (Simonyan et al., 2014), Integrated Gradients (IG) (Sundararajan et al., 2017), and GradientSHAP (Lundberg and Lee, 2017). *Perturbation* methods: LIME (Ribeiro et al., 2016), Feature Ablation, and RISE (Petsiuk et al., 2018) mask the input and calculate importance from the output change.

RISE is our default explainer because it is black-box and aligns naturally with the perturbation philosophy of the causal test. It samples N low-resolution binary masks ($s \times s$ grid, on-probability p_1), shifts them to image size, multiplies each with x , runs the model, and estimates importance as the probability weighted average of the masks: $S = \frac{1}{N p_1} \sum_m f_c(x \odot m) m$. We use $N=5000$, $s=10$, $p_1=0.1$ for ResNet50 and $N=2000$ for the other models.

2.2 The causal insertion/deletion pipeline

Given S , pixels are sorted by importance. Deletion replaces the top ranked pixels with a baseline (black) in steps and records f_c . A good map shows a fast drop and low AUC. Insertion starts from a Gaussian blur baseline and restores top pixels. A good map yields a fast rise and high AUC. We compute the AUC of the confidence vs fraction curve, and two stability statistics: the Spearman rank correlation between perturbation level and confidence (ideal -1 for deletion, $+1$ for insertion) and a *monotonicity ratio* (fraction of steps moving in the expected direction). Higher monotonicity ratio is better.

2.3 ReCalX calibration

Heavily masked images lie far from the training distribution, so f 's confidence becomes unreliable and the causal curve is biased and noisy. Decker et al. (2025) formalise this link between uncertainty calibration and perturbation-based explanation quality and propose *ReCalX*: recalibrate the model for the specific perturbations used, without altering its predictions. We adopt a binned temperature-scaling realisation. The perturbation level $\ell \in [0, 1]$ is split into $B=10$ bins; for each bin b we collect raw logits over a calibration set and fit a scalar temperature T_b by minimising cross-entropy with L-BFGS-B. At evaluation, the wrapper `ReCalXModel` selects the bin for the current step and returns z/T_b (softmax applied afterwards). Because $\arg \max_c z_c$ is invariant to dividing by $T_b > 0$, calibrated and raw models make identical predictions; only the confidence is corrected.

Calibration metrics. We measure miscalibration with two complementary metrics. *Expected Calibration Error* (ECE) (Guo et al., 2017) partitions confidence into 10 equal-width bins and reports the sample-weighted average $|\text{acc}(B_b) - \text{conf}(B_b)|$; perfect calibration gives $\text{ECE} = 0$. *Thresholded Adaptive Calibration Error* (TACE) (Nixon et al., 2019) uses 15 equal-count bins after discarding steps with confidence < 0.01 , making it more robust for deletion sequences where many steps produce near-uniform outputs. Both metrics better when lower.

2.4 Bucketing

The classic approach removes a fixed top- $n\%$ (e.g. 1%) per step, which can make the derived curve overly noisy. *Significance bucketing* instead sorts pixels by rank and removes N equal-count

groups, one per step. We use $N=25$ buckets for evaluation and contrast against fixed-step 1% deletion.

2.5 Datasets and models

We use an ImageNet validation subset (20 images sampled per run, seed 42). Models are torchvision ImageNet pretrained ResNet50, ResNet101 (He et al., 2016), ViT-B/16 and ViT-B/32 (Dosovitskiy et al., 2021).

3. Experiments

All runs use $N=20$ images. R Calibration temperatures are fit in a batch unless stated otherwise. All reported p -values denote the Wilcoxon signed-rank test.

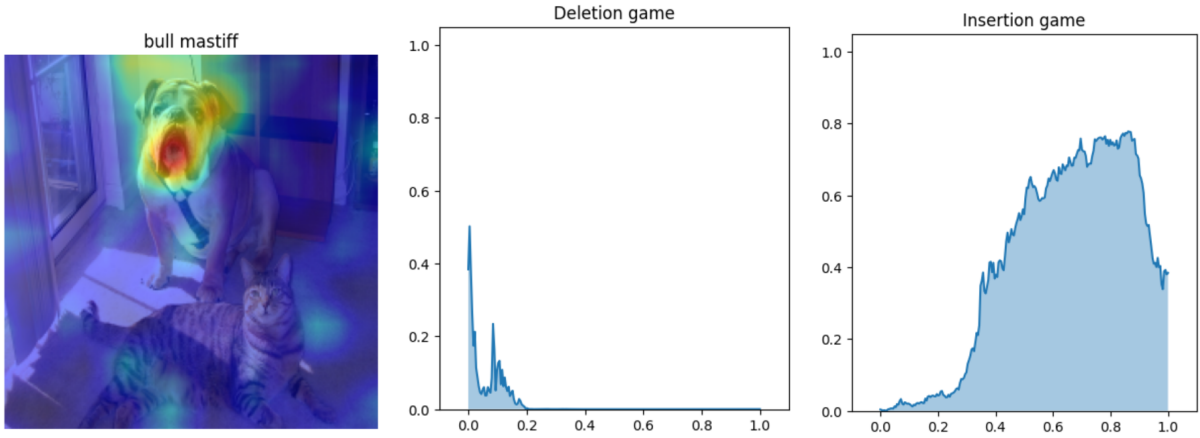


Figure 1: Example saliency map (ResNet50 with RISE) with its deletion/insertion curves.

3.1 Calibrated causal analysis

The fitted per-bin temperatures (Table 5 in the Appendix) reveal different model behaviours. ResNet50 requires $T < 1$ at nine of ten bins, indicating systematic underconfidence on perturbed images; only the 0.9–1.0 bin shows $T = 1.37$. ViT-B/16 temperatures exceed 1 from the 0.5–0.6 bin onwards and climb steeply to $T = 5.0$ at 90–100% masking, showing progressive overconfidence at high perturbation levels.

Table 1 shows ECE and TACE for both architectures. Calibration reduces both metrics for 5 of 6 methods on both models; the exception is IntGrad ECE on ResNet50.

Method	ECE (raw / cal)		TACE (raw / cal)	
	ResNet50	ViT-B/16	ResNet50	ViT-B/16
RISE	0.280 / 0.170	0.204 / 0.131	0.355 / 0.177	0.229 / 0.195
IntGrad	0.163 / 0.165	0.308 / 0.122	0.284 / 0.239	0.356 / 0.211
LIME	0.422 / 0.094	0.259 / 0.150	0.434 / 0.086	0.274 / 0.161
GradShap	0.186 / 0.176	0.274 / 0.127	0.283 / 0.236	0.324 / 0.203
Saliency	0.211 / 0.168	0.210 / 0.146	0.300 / 0.221	0.220 / 0.180
FeatAbl.	0.390 / 0.205	0.255 / 0.182	0.452 / 0.225	0.268 / 0.205

Table 1: ECE and TACE for ResNet50 and ViT-B/16 (raw / calibrated). All improvements are significant (Wilcoxon $p < 0.001$ except IntGrad ECE on ResNet50). Results in mean ECE $0.28 \rightarrow 0.16$, mean TACE $0.35 \rightarrow 0.20$ for ResNet50.

Table 2 reports the effect on curve shape. Calibration significantly improves Spearman r for RISE, LIME, and FeatAblation, and monotonicity for LIME and FeatAblation ($p < 0.01$). IntGrad monotonicity significantly worsens ($p = 0.017$); gradient methods with already strong raw Spearman r (GradShap: -0.925 , Saliency: -0.891) show no significant change.

Method	Spearman r (ideal -1)			Monotonicity (ideal 1)		
	raw	cal	p	raw	cal	p
RISE	-0.811	-0.909	$.033^*$	0.632	0.634	$.913$
IntGrad	-0.934	-0.860	$.486$	0.644	0.590	$.017^*$
LIME	-0.588	-0.920	$< .001^*$	0.562	0.638	$< .001^*$
GradShap	-0.925	-0.837	$.170$	0.646	0.602	$.069$
Saliency	-0.891	-0.831	$.674$	0.648	0.634	$.742$
FeatAbl.	-0.483	-0.796	$.006^*$	0.580	0.670	$< .001^*$

Table 2: Effect of ReCalX on Spearman r and monotonicity ratio for ResNet50; * means Wilcoxon $p < 0.05$.

3.2 Bucketing vs fixed step

Bucketing significantly outperforms fixed-step 1% deletion on both Spearman and monotonicity ratio, for both raw and calibrated model.

Schedule	Spear _{raw}	Spear _{cal}	Mono _{raw}	Mono _{cal}
TopN (1%)	-0.754	-0.879	0.555	0.578
Bucket (25)	-0.811	-0.906	0.632	0.664

Table 3: Bucketing (25 groups) vs TopN (1%) on ResNet50 with RISE. Bucket outperforms TopN on both metrics with Wilcoxon $p < 0.01$ for Spearman and monotonicity.)

4. Conclusions

The insertion/deletion reliability gap is a calibration issue: perturbed images are OOD and the raw model becomes miscalibrated. *ReCalX* addresses this without altering predictions, significantly reducing ECE and TACE for five of six attribution methods. Bucketing further stabilises the test, significantly outperforming fixed-step removal on both Spearman r and monotonicity. The calibration benefit holds across all four tested architectures (Table 4 in the Appendix).

The effect of calibration on curve *monotonicity* is method-dependent: it significantly improves LIME and Feature Ablation but can worsen monotonicity for strong gradient methods (IntGrad, GradShap). Resolving the exact mechanism behind this divergence remains an open question.

Limitations and future work. The primary limitation of this study is its evaluation scale ($N = 20$ images per architecture) and its restriction to image classification. The ReCalX framework also introduces a slight computational overhead by requiring batch-fitted temperatures. Promising extensions include investigating the aforementioned gradient-method divergence, aligning deletion steps with ViT tokenisation, and using FID or OTDD distances to rigorously quantify how far specific perturbations push inputs out of distribution.

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Appendix

Architecture	Spear _{raw}	Spear _{cal}
ResNet50	−0.900	− 0.936
ResNet101	−0.879	− 0.909
ViT-B/16	−0.909	− 0.935
ViT-B/32	−0.954	− 0.969

Table 4: Spearman r (raw and calibrated) for all four architectures, using Integrated Gradients and BucketDeletion(25), $N=20$. Calibration consistently shifts r toward -1 across both families (ResNet and ViT), confirming the finding is not architecture-specific.

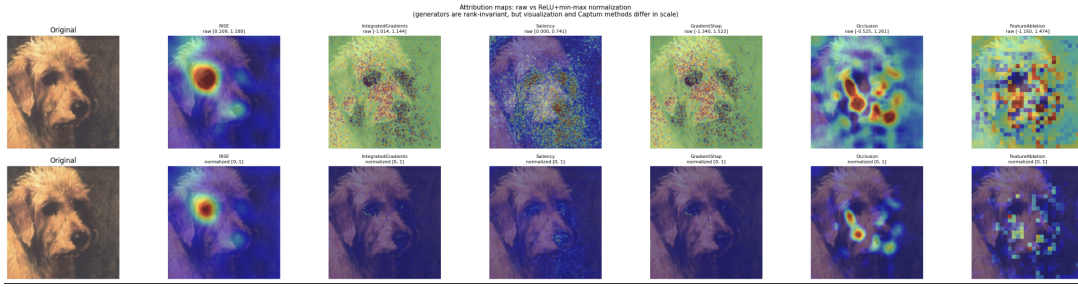


Figure 2: Baseline visualizations of the evaluated attribution methods. These initial importance rankings dictate the pixel removal order during the causal deletion tests.

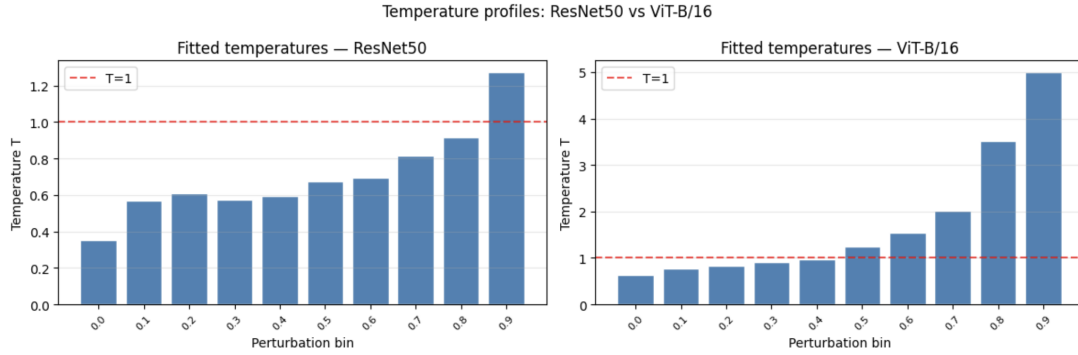


Figure 3: Fitted temperature profiles across perturbation bins for ResNet50 and ViT-B/16. This visualizes the data from Table 5, highlighting the varying OOD response: ResNet50 is primarily underconfident ($T < 1$), while ViT-B/16 becomes highly overconfident at heavy masking levels.

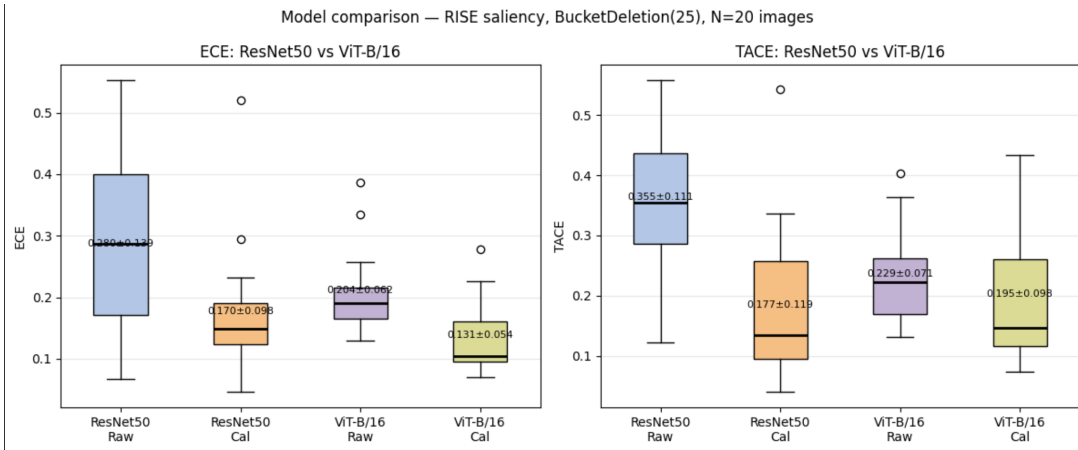


Figure 4: Raw versus calibrated ECE and TACE using BucketDeletion. The charts demonstrate ReCalX substantially closing the reliability gap, cutting miscalibration roughly in half across both architectures.

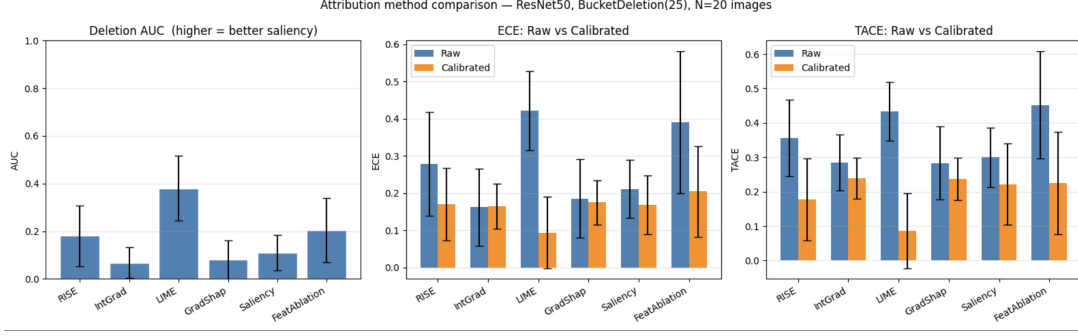


Figure 5: Attribution method comparison on ResNet50 (BucketDeletion(25), $N=20$ images). The left panel shows the baseline Deletion AUC for six attribution methods. The center and right panels illustrate the impact of recalibration, demonstrating that perturbation-aware temperature scaling significantly reduces both ECE and TACE across almost all tested methods, thereby closing the OOD reliability gap.

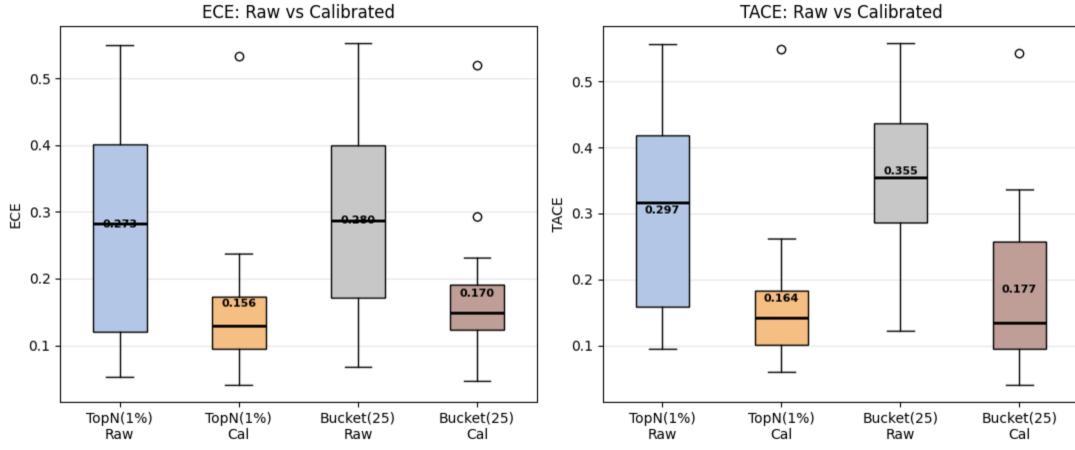


Figure 6: Comparison of ECE and TACE using TopN(1%) versus Bucket(25) removal schedules on ResNet50. The boxplots demonstrate that both fixed-step removal and significance bucketing suffer from high initial miscalibration (Raw), and both see their median errors drastically reduced after post-hoc temperature scaling (Cal).

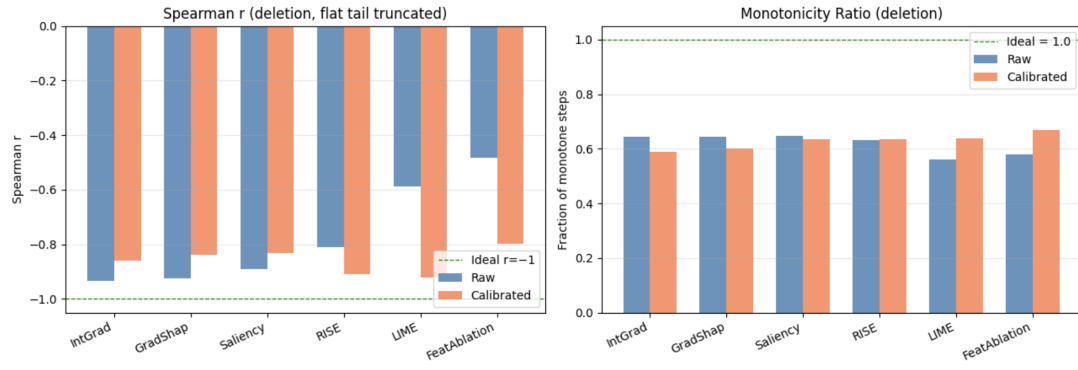


Figure 7: Spearman r and monotonicity ratio distributions (raw vs. calibrated) for RISE on ResNet50. Calibration pushes the rank correlation closer to perfect monotonicity (-1). But only for part of all attribution maps.

Model	Perturbation bin (fraction masked)									
	0–.1	.1–.2	.2–.3	.3–.4	.4–.5	.5–.6	.6–.7	.7–.8	.8–.9	.9–1
ResNet50	0.24	0.52	0.57	0.55	0.60	0.67	0.70	0.79	0.96	1.37
ViT-B/16	0.62	0.73	0.81	0.88	0.97	1.18	1.52	2.04	3.46	5.00

Table 5: Batch-fitted temperature T_b per perturbation bin (RISE, BucketDeletion(25), $N=20$). $T<1$ amplifies confidence (model underconfident); $T>1$ softens it (overconfident). **Bold** = bins requiring softening. ResNet50 is underconfident at all but the last bin; ViT-B/16 becomes progressively overconfident from 50% masking onward, reaching $T=5.0$ at 90–100%.

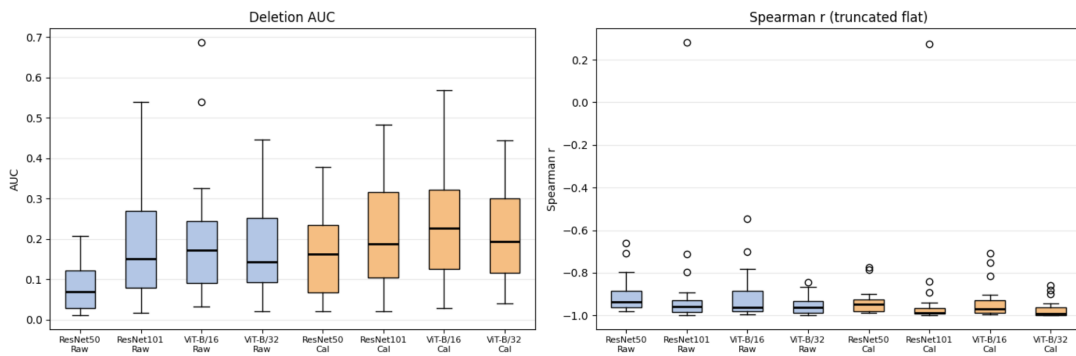


Figure 8: Cross-architecture comparison of raw versus calibrated AUC and Spearman r .